

THE JOURNEY · 2026

# Hotel Pipeline

*From scattered PDFs to a production data platform.*

0 → 100

WHAT'S INSIDE

Where we started

Six phases of work

Where we landed

What comes next

---

REVENUE INTELLIGENCE TEAM

**00**  
**—**

# Seven hotels. Stacks of PDFs. No single source of truth.

## RS02, RS03, ST14 reports

Three different PDF formats per hotel, every day.

## Manual aggregation

Someone opens each PDF and types into a spreadsheet.

## Numbers never align

Different totals depending on who pulled them.

## No history kept

Yesterday's snapshot is gone the moment today's run starts.



# 01 Built the parsers.

*Three Python modules that read every PDF format, every layout, every edge case.*

| RS02                                                                                                                                                                                                | RS03                                                                                                                                                                                     | ST14                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><h2>Reservations parser</h2><p>Extracts every booking — rsvn_no, dates, rooms, pax, rate, revenue, agent, market.</p><hr/><p><i>Multi-room aggregation · Document-date detection</i></p></div> | <div><h2>Cancellations parser</h2><p>Captures every cancelled reservation with cancel date, lead days, lost revenue.</p><hr/><p><i>Format-agnostic across all 7 properties</i></p></div> | <div><h2>Market on-book parser</h2><p>Daily on-book snapshot by stay-date — rooms reserved, revenue, market segment.</p><hr/><p><i>Source of truth for occupancy + RevPAR</i></p></div> |

*Result: every PDF becomes structured rows, automatically.*

# 02 All parsed rows land in DuckDB.

*Three raw fact tables. Daily-snapshot primary keys. One file, queryable like Postgres.*



## fact\_reservations

PK: (rsvn\_no, hotel\_name, report\_date)

Atomic booking facts. Same booking on different report dates = different rows (snapshot model).



## fact\_cancellations

PK: (rsvn\_no, hotel\_name, report\_date)

Every cancellation, with cancel reason, lead days, lost revenue.



## fact\_market\_onbook

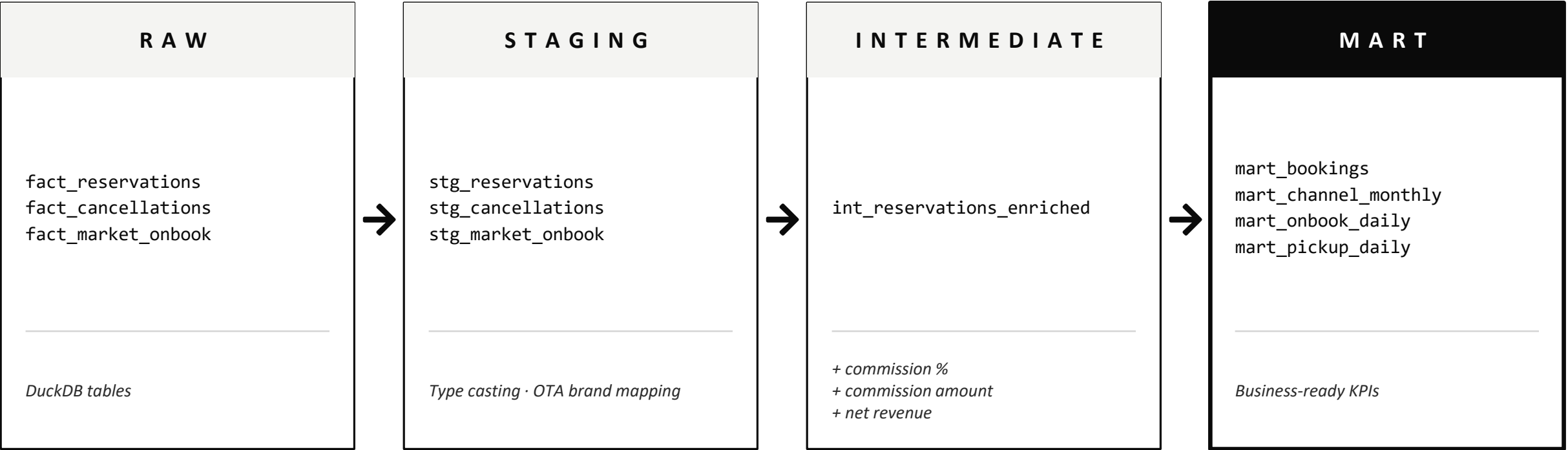
PK: (hotel\_name, stay\_date, market, report\_date)

Daily on-book by stay-date and market — drives occupancy, RevPAR, pace.

*One DuckDB file. Six months of history. Queryable from any SQL client.*

# 03 Layered analytics with dbt.

*Three layers. Each layer cleaner than the one below.*



*Every layer documented, tested, and rebuilt nightly.*

# 04 OTA commission, calculated — not guessed.

*Per OTA × per hotel commission table. Net revenue reconciles exactly.*

THE FORMULA

**net\_revenue**

$$\begin{aligned} &= \text{gross\_revenue} \\ &- (\text{gross\_revenue} \times \text{commission\_pct}) \end{aligned}$$

---

$$\text{commission\_pct} = \text{ota\_brand} \times \text{hotel\_name}$$

*70 rows — 10 OTA brands × 7 hotels*

*Rates negotiated by each property.*

**10**

OTA BRANDS MAPPED

*Agoda · Booking · Expedia · Ctrip · ...*

**100%**

OTA COMMISSION COVERAGE

*Every OTA booking has a rate*

**฿50.9M**

COMMISSION SURFACED

*Over 6 months across all hotels*

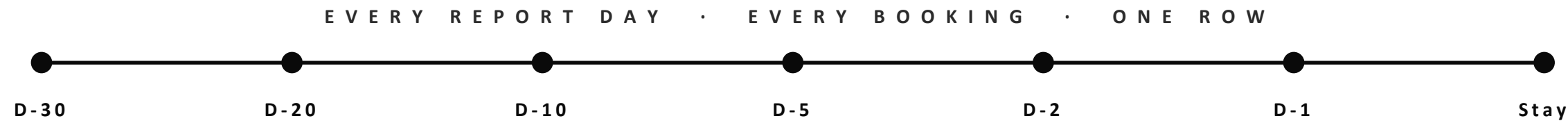
**฿289.2M**

NET REVENUE COMPUTED

*Gross less commission, reconciled*

# 05 Daily snapshots — pace, pickup, history.

*Same booking captured on every report day → we see how forecasts evolve.*



## **mart\_bookings**

Atomic booking fact, deduped to first-seen — true production date.

## **mart\_channel\_monthly**

OTA scorecard: bookings, commission %, ADR by hotel × month.

## **mart\_onbook\_daily**

Occupancy + RevPAR by stay-date using latest snapshot.

## **mart\_pickup\_daily**

LAG over snapshots → daily pickup of rooms and revenue.

*Pace and pickup are now first-class — not derived from screenshots.*

# 06

## Validated, documented, queryable.

*Every number traceable. Every table self-describing. Every model tested.*

### VALIDATION

- Hotel-by-hotel row-count match vs PDF grand totals
- Revenue reconciles within \$1 per file
- Room capacity cross-checked against external ADR report
- 100% OTA commission coverage validated

### DOCUMENTATION

- Model + column descriptions written in dbt
- Persisted as COMMENT ON in DuckDB
- Visible in DBeaver and any SQL client
- PROJECT\_REPORT.md — end-to-end narrative

### OPERATIONS

- Idempotent loads — re-running is safe
- Snapshot model preserves daily history
- Single-command refresh: run\_pipeline.bat
- All sources under version control (git)



# Where we landed.

*From scattered PDFs to a production data platform.*

# 100

---

7

HOTELS LIVE

53,740

BOOKINGS PARSED

\$340M

REVENUE TRACKED

\$50.9M

COMMISSION VISIBLE

4

BUSINESS MARTS

180+

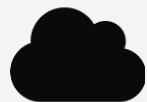
DAYS OF HISTORY

*Foundation complete. Platform proven. Numbers reconciled.*



# From running it to operating it.

*Foundation is done. Next: make it always-on and team-accessible.*



## Cloud warehouse

Move DuckDB → cloud. Multi-user. Auto-backup. Mobile + tablet access. Shared URL for the whole team.

PHASE 02 · PROPOSED



## Airflow orchestration

Self-running daily. Auto-retry on failure. Slack alerts. One-click backfill. Full audit trail.

PHASE 02 · PROPOSED

***The data is here. The platform is the next chapter.***